
CIS*4820 Assignment 2

1. NPC Stats

General Information

- **Model**: Provided starter NPC model (or custom model if applicable).
- **Animations**: Idle, Walk, Run, Attack, Die.
- **Health**:
    - Maximum Health: 100
    - Starting Health: 100
    - Health Threshold for Running Away: 10% (when health < 10%).
- **Movement Speed**:
    - Walk Speed: 3 units/second.
    - Run Speed: 6 units/second.
- **Sight Radius**: 15 units.
- **Field of View**: 180° (this can be edited through the Unity editor).
- **Attack Range**: 2 units.
- **Run Away Distance**: 10 units (to start fleeing when health is <10%).
- **Damage dealt**: 10 damage per attack (this can be adjusted as needed).

AI Behaviour Summary

The NPC has several core states:

- **Idle**: When the player is not detected.
- **Walk**: When the player is within the NPC's sight radius and within the walk range.
- **Run**: When the player is within the sight radius but beyond the walk range.

- **Attack**: When the NPC is within attack range of the player.
- **Run Away**: When the NPC's health is less than 10% and the player is detected.
- **Die**: When the NPC's health reaches 0.

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2. State Machine Diagram

The state machine for the AI is structured as follows:

States

1. **Idle State**:
    - **Transition**: The NPC stays in this state when the player is not within the sight radius or is not detected.
    - **Action**: Plays the idle animation.
    - **Exit Condition**: The player enters the sight radius.
2. **Walk State**:
    - **Transition**: The NPC walks towards the player when the player is within the sight radius and within walking range.
    - **Action**: Plays the walking animation.
    - **Exit Condition**: The player enters attack range or moves out of sight radius.
3. **Run State**:
    - **Transition**: The NPC runs towards the player if the player is visible but outside the walking range (within sight radius but beyond walking range).
    - **Action**: Plays the running animation.
    - **Exit Condition**: The player enters attack range.
4. **Attack State**:
    - **Transition**: The NPC attacks the player when within attack range.

- **Action**: Applies damage to the player, plays the attack animation.
- **Exit Condition**: Player is hit or health reaches 0 (NPC dies).

5. **Run Away State**:
    - **Transition**: The NPC transitions to this state if health is below 10% and the player is detected.
    - **Action**: Plays the run animation (moves away from the player).
    - **Exit Condition**: NPC's health goes above 10% or NPC escapes a safe distance.
6. **Die State**:
    - **Transition**: The NPC transitions to this state if health reaches 0.
    - **Action**: Plays the die animation, deactivates the NPC object.
    - **Exit Condition**: None (permanent state).

State Transitions

From State	To State	Condition/Trigger
Idle	Walk	Player enters sight radius (within walk range)
Idle	Run	Player enters sight radius (outside walk range)
Walk	Attack	Player enters attack range (within walk range)
Walk	Run	Player moves out of walk range but within sight radius
Run	Attack	Player enters attack range
Run	Run Away	NPC health < 10% and player detected
Run Away	Run	NPC health > 10% or player is no longer in sight radius
Attack	Idle	Player moves out of attack range or NPC health reaches 0
Attack	Walk/Run	NPC health > 10% and player moves out of attack range
Die	None	NPC health = 0

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3. Enhancements

Basic Enhancements

- **Both Player and NPC Damage**: The player and the AI now apply damage to each other in discrete increments per swing/attack. This ensures the player must perform multiple attacks to deal damage, adding more depth to the combat mechanics.
- **Power-up on Kill**: When an NPC is killed, the player receives a power-up: +10 health. This is achieved by updating the player's health whenever an NPC dies.
    - **Power-up Logic**: Once an NPC's health drops to 0, the AddHealth method is called to increase the player's health by 10.

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4. Conclusion

This documentation outlines the fundamental AI behavior implemented in the assignment. The AI uses a finite state machine to transition between different states based on the player's position and the NPC's health. Several enhancements, such as discrete damage application, health power-ups for the player, and the potential for adding new enemy types, have been proposed.

For the best results, it's important to refine the AI's state transitions to ensure smooth and responsive behavior. Additionally, fine-tuning movement and attack mechanics will make combat feel more dynamic and engaging.
